

BLOCK 4
SECTOR SPECIFIC ISSUES
AND POLICIES

BLOCK 4 SECTOR SPECIFIC ISSUES AND POLICIES

This block focuses on Sector-specific (Agriculture, Industries and Services) issues, concerns and associated policies addressing those issues and concerns. The block comprises 5 Units.

Unit 13 entitled ‘**Agriculture: Issues, Concerns, Policy and Programmatic Initiatives**’ deals with role and relevance of agricultural sector in the Indian economy, trends of agricultural production, causes of deceleration of agricultural growth in India, shifting of agriculture from traditional to modern dynamic sector, impact of green revolution and the various policy initiatives taken by the government in this regard.

Unit 14 on **Large Scale Industries in India: Issues and Policy** covers the issues and policy related to large scale Industries. The unit examines growth strategy of heavy industries in terms of their features, legislative support, industrial policy support, critical issues before industrial sector and the framework for a new industrial policy.

Unit 15 on ‘**Micro, Small and Medium Enterprises: Issues and Policy**’ covers issues and challenges being encountered by MSME sector, the impact of demonetization and GST on the MSME sector, and the various policy initiatives taken by the Government.

Unit 16 entitled ‘**Services Sector I: Organised Sector— Issues and Policy**’ identifies the various issues and concerns of organised services sector. Sector-specific policy initiatives and issues in selected organised sectors have been provided in this unit.

Unit 17 entitled ‘**Services Sector-II: Informal Sector— Issues and Policy**’ sheds light on salient characteristics of informal service sector. Factors responsible for its exponential growth, the issues and challenges being faced by the informal sector, policy implications of informality, and the role of government at various levels in this regard have also been covered in this Unit.

UNIT13 AGRICULTURE: ISSUES, CONCERNS, POLICY AND PROGRAMMATIC INITIATIVES

Structure

- 13.0 Objectives
- 13.1 Introduction: Role and Relevance of Agriculture in the Indian Economy
- 13.2 Agriculture Production and Productivity after Independence
 - 13.2.1 Trends in Production
 - 13.2.2 Trends in Productivity
- 13.3 Causes for Stagnation in Agriculture Growth in India
 - 13.3.1 Deceleration in Agriculture Investment
 - 13.3.2 Inadequate Irrigation Coverage and Falling Productivity of Irrigation in Food grains
 - 13.3.3 Failure to Evolve and Adopt New Technologies
 - 13.3.4 Shrinking Farm Size
 - 13.3.5 Non-availability or Inadequate Credit from Formal Sources
 - 13.3.6 Imbalance in the Use of Fertilizers
 - 13.3.7 Effectiveness of Minimum Support Prices (MSPs)
 - 13.3.8 Inadequate Availability and Access to Quality Seeds
 - 13.3.9 Farm Mechanisation: Major Challenges
 - 13.3.10 Post-harvest Activities-Inadequate Storage Facilities
- 13.4 Transformation of Indian Agriculture: Strategies for Development (1951-2002)
 - 13.4.1 Phase I: Traditional Agriculture (1951-1966)
 - 13.4.2 Phase II: Technologically Dynamic Agriculture with New Agriculture Strategy (NAS):1966 – 2002
 - 13.4.3 Impact of Green Revolution on Production and Yield of Major Crops During the Entire Planning Period
- 13.5 Transformation of Indian Agriculture: Strategies for Development (2002-2014)
 - 13.5.1 Rashtriya Krishi Vikas Yojana (RKVY)
 - 13.5.2 Second Green Resolution
- 13.6 Transformation of Indian Agriculture through an Umbrella Programme of Doubling of Farmers' Income (DFI) from 2015 to 2022
 - 13.6.1 Prime Minister's Multi-dimensional Seven Points Strategy
 - 13.6.2 Policy Initiatives for Enhancing the Farmers' Output per Hectare: Improving Cropping Intensity
 - 13.6.3 Policy Initiatives for Crop Diversification and Enhancing Value Realisation
 - 13.6.4 Policy Initiatives to Ensure Remunerative Price to Farmers for their Produce: MSP Policy and Procurement
 - 13.6.5 Policy Initiatives to Improve Resource Efficiency/ Reduction in Cost
- 13.7 Relevance of Non-Agricultural Activities in Doubling of Farmers' Income
- 13.8 Let Us Sum Up
- 13.9 Term-end Exercises

13.0 OBJECTIVES

After reading and studying this unit, you will be able to explain:

- role and relevance of agriculture sector in the Indian Economy;
- trends in agriculture production and productivity both in major food grains and non-food grains since independence especially after the adoption of New Agriculture Strategy in mid-1960s;
- causes for Stagnation in Agriculture Growth in India especially during economic reforms regime;
- shift in agricultural sector in India from a traditional agriculture in the 1950s to the modern technologically dynamic high capital-intensive agriculture;
- various programmes implemented during the first decade after independence for increasing the food grains production;
- impact of green revolution in terms of attaining self-sufficiency in food grains and in terms of shrinking incomes of the farmers as well as widespread environmental implications; and
- various policy and strategic initiatives by the Government for doubling of farmers' income.

13.1 INTRODUCTION: ROLE AND RELEVANCE OF AGRICULTURE IN INDIAN ECONOMY

Agriculture as a sector has contributed significantly in the process of India's development in number of ways, some of which are listed below:

- *Contribution to National Income:* In 1950-51 contribution of agricultural sector to national income was 59 per cent and in 2004-05 it came down to 24.4 per cent and further to about 15 per cent in 2017-18. Economic contribution of agriculture to India's GDP is steadily declining with the country's broad-based economic growth, yet, nearly 43 per cent of the population continues to be dependent on it for livelihood.
- *Largest Employment Provider Sector:* In 1972-73, 73.9 per cent of the working population was engaged in agriculture/allied activities, which fell to 64.8 per cent in 1993-94, to 48.9 per cent in 2011-12 and to 43 per cent in 2017-18. With rapid increase in population, the absolute number of people engaged in agriculture has become exceedingly large. As development of other sectors has not been sufficient, dependence of increasing population on agriculture has increased with marginal productivity of labour being less than or equal to zero— a scenario which gives rise to the familiar problem of disguised unemployment in agriculture.
- *Provision of Food Surplus to the Expanding Population:* Agriculture is the basic source of food supply and if agriculture fails to meet the rising demand of food products (due to increase in population), it is bound to affect adversely the growth rate of the economy. In India, food consumption still constitutes

on an average 60 per cent of household income thereby generating adequate demand for food products.

- *Providing Raw Materials to Industries:* It provides essential raw materials to many industries like cotton textiles, jute, sugar, vegetables, oil, tinned food, Cigarettes and rubber, etc. The entire range of food processing industries is also dependent on agriculture. Therefore, unless agriculture develops, these industries will not be able meet their demand for raw materials.
- *Market for Industrial Products:* As majority of the population live in rural areas, increased rural purchasing power is a valuable stimulus to industrial development, as propounded by Ragner Nurkse and Arthur Lewis in their theories on financing industrial development through agriculture
- *Commercial Importance:* For a number of years after independence, the three agriculture-based exports of India – cotton textiles, jute and tea – accounted for more than 50 per cent of export earnings of the country. On adding the exports of other agricultural products such as cashew kernels, tobacco, coffee, Vanaspati, oil, and sugar, etc. the share of agriculture in total exports rose to around 70 to 75 per cent. Such a heavy dependence on agriculture reflected the underdeveloped nature of economy. With economic progress and consequent diversification of production base, share of agriculture goods in total exports fell to 44.2 per cent in 1960-61 which considerably fell to 30.7 per cent in 1980-81 and 12.3 per cent in 2016-17.

As India is still home to the largest number of poor and malnourished people in the world with it being at the 101th position among 119 countries in Global Hunger Index, 2018, a higher priority to agriculture will achieve the goals of reducing poverty and malnutrition as well as achieving inclusive growth. 'Growth with inclusiveness' can be achieved only when agriculture growth accelerates and is widely shared among the people and regions of the country.

Given the role and relevance of agriculture in India's economy, this unit intends to discuss in detail (a) the performance of agriculture sector in terms of trends in production and productivity focusing upon the period since independence; (b) Issues and concerns being faced by the sector; and (c) Strategic, policy and schematic initiatives taken by the government for transformation of agriculture and how far these measures have benefited in *de-facto* sense the farmers/cultivators.

13.2 AGRICULTURE PRODUCTION AND PRODUCTIVITY AFTER INDEPENDENCE

The two major crop groups are Food grains (Food crops) and Non-food grains (Non-food crops). Food crops include rice, wheat, jowar, bajra, small millets, cereals, gram, pulses, vegetables and fruits, while Non-food crops include Oilseeds (groundnut, linseed, rapeseed and mustard, niger seed), Fibre crops (cotton, jute and mesta), sugarcane, tobacco, Plantation crops (tea, coffee and rubber), Condiments and Spices.

The food crops contribute about 2/3rd of total agricultural production. The most important component in the food grains category is wheat followed by rice. In

non-food grains category, oilseeds constitute the most important group followed by Sugarcane and cotton.

13.2.1 Trends in Production

Table 13.1 reveals the following trends:

- Consistent increase in food grains production with considerable annual fluctuations.
- The production of two major crops – Wheat and Rice – has consistently increased since 1950-51 and increase was significant, especially in case of wheat after the adoption of New Agricultural Strategy since mid-1960s and in rice it followed subsequently in early 1980s. These two crops benefitted the most from the development of ‘miracle seeds’.
- Jowar and bajra exhibited erratic trends over the planning period. Introduction of HYV maize has pushed up its production and productivity. As a result, area under maize has risen significantly at the cost of other cereals like jowar, bajra, etc.

Table 13.1: Trends in Agricultural Production 1950-51 to 2018-19 (Units in Million Tonnes)

Crop	Decadal Trends from 1950-51 to 2010-11 and Annual Trends thereafter up to 2018-19														
	1950-51	1960-61	1970-71	1980-81	1990-91	2000-01	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Rice	20.6	34.6	42.2	53.6	74.3	85.0	96.0	105.3	105.2	106.6	105.5	104.4	109.7	112.8	116.4
Wheat	6.4	11.0	23.8	36.3	55.1	69.7	86.9	93.5	93.5	95.8	86.5	92.3	98.5	99.9	102.6
Jowar	5.5	9.8	8.1	10.4	11.7	7.5	7.0	6.0	5.3	5.5	5.5	4.2	4.6	4.8	3.8
Bajra	2.6	3.3	8.0	5.3	6.9	6.8	10.4	10.3	8.7	9.3	9.2	8.1	9.7	9.2	8.6
Maize	1.7	4.1	7.5	7.0	9.0	12.0	21.7	21.8	22.3	24.3	24.2	22.6	25.9	28.8	27.2
Other Cereals	4.1	6.5	7.0	6.4	4.1	4.5	4.3	5.3	3.6	4.2	3.9	3.6	3.4	6.4	3.0
Pulses	8.4	12.7	11.2	10.6	14.3	11.0	18.2	17.1	18.3	19.3	17.2	16.4	23.1	25.4	23.4
Total FGs	50.8	82.0	108.4	129.6	175.4	196.8	244.5	259.3	257.1	265.0	252.0	251.6	275.1	285.0	285.0
Oilseeds#	6.2	7.0	9.6	9.4	18.6	18.4	32.5	29.8	30.9	32.7	27.5	25.3	31.3	31.5	32.3
Sugarcane	57.1	110.0	126.4	154.2	241.0	296	342.4	361.0	341.2	352.1	362.3	348.4	306.1	379.9	400.2
Cotton@	3.0	5.6	4.8	7.0	9.8	9.5	33.0	35.2	34.2	35.9	34.8	30.0	32.6	32.8	27.8
Jute\$	3.3	4.1	4.9	6.5	7.9	9.3	10.0	10.7	10.3	11.7	10.6	9.9	10.4	9.6	9.4

@Million Bales (bale=170kg), \$ Million Bales (bale=180kg),

#Includes groundnut, rapeseed & mustard, sesamum, linseed, castorseed, nigerseed, safflower, sunflower and soyabean;

Source: Source: Indian Economy, Mishra and Puri, 36th Edition, 2018 and Col 3-6 from Economic Survey, 2000-01 (Statistical Tables, Table 1.12), Col 7 to 16 from Economic Survey, 2019-20 (Volume II), Statistical Appendices, Table 1.15, page number A 34.

- Pulses requirement in India is estimated at about 17 million tons; but actual production has been considerably less than this level during most of the period. However, production of pulses shot up from 16.4 million tons in 2015-16 to 23 million tons in 2016-17. The supply of pulses lagged behind for most of the planning era; India has had to import a large quantity of pulses over the period which constituted 15-20 per cent of our total demand in the country. Recently India has achieved nearly self-sufficiency in pulses

and therefore imports are confined to international agreements negotiated earlier

- As far non food grains group is concerned, the production of oilseeds rose sharply in the latter half of 1980s and the first decade of the present century with minor fluctuations. Just like pulses, there has been a large gap between demand and supply for a considerable period forcing the country to import large quantities of edible oils. In fact, even at present 60 per cent of demand for edible oils is met out of imports, Which is likely to be untenable in the long run.
- century with minor fluctuations. Just like pulses, there has been a large gap between demand and supply for a considerable period forcing the country to import large quantities of edible oils. In fact, even at present 60 per cent of demand for edible oils is met out of imports, which is likely to be untenable in the long run.
- Production of cotton which averaged 8 to 9 million bales per annum till 2000-01 rose significantly thereafter due to widespread use of Bt Cotton (a genetically modified pest resistant plant cotton variety) in 2002. From annual production of about 9 million bales in 2000-01, the production rose to 33 million bales in 2010-11 but thereafter it almost stagnated at the same level. Now almost 90 per cent of cotton area is covered under Bt cotton.
- Jute saw a slow and halting progress during the entire period of planning.
- The sugarcane exhibited a consistent increase in production during the four decades period 1952-53 to 2002-03 but thereafter it rose considerably with annual fluctuations.

13.2.2 Trends in Productivity

The agricultural production depends not only on the area but also on the productivity of land. It shows the relationship between inputs and output. The agricultural productivity can be classified into two categories viz. agricultural productivity per hectare and agricultural productivity per worker. Trends in productivity per hectare of major food grains and Non-food grains crops after independence are analysed in Table 13.2.

The Table 13.2 reveals the following trends:

- The yield per hectare of all food grains has increased by about four times from 552 kg per hectare in 1950-51 to 2299 kg per hectare in 2018-19.
- The most significant increase has been recorded by wheat with its yield increasing from 655 kg per hectare to 3507 kg per hectare in 2018-19. Productivity of rice also increased significantly from 1123 kg per hectare in 1970-71 to 2659 kg per hectare in 2018-19.
- Jowar and Bajra recorded much lower rates of growth in productivity.
- Most disappointing has been the performance of pulses with stagnant productivity at around 532 kg per hectare during the period from 1960-61 to 1999-00, thereafter it increased at snail pace to 806 in 2018-19.
- The productivity of maize and cotton has increased in recent years due to adoption of hybrid maize varieties and Bt cotton since 2002 in most areas of the country.

Table-13.2: Yield Per Hectare of Major Crops (Kg per Hectare)

Crop	1950-51	1960-61	1970-71	1980-81	1990-91	2000-01	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Rice	668	1013	1123	1336	1740	1901	2239	2393	2462	2416	2391	2400	2416	2576	2659
Wheat	655	851	1307	1630	2281	2708	2989	3178	3117	3145	2750	3034	3145	3368	3507
Jowar	353	533	466	660	814	764	949	957	850	957	884	697	957	960	979
Bajra	288	286	622	458	658	688	1079	1171	1198	1184	1255	1132	1184	1231	1242
Maize	547	926	1269	1159	1518	1822	2540	2478	2566	2676	2632	2563	2676	3065	2986
Pulses	441	539	524	473	578	544	691	699	789	764	728	656	764	853	806
Total FGs	552	710	872	1023	1380	1626	1930	2078	2079	2120	2028	2042	2120	2235	2299
Oilseeds*	481	507	579	532	771	810	1193	1133	1168	1168	1075	968	1168	1284	1265
Cotton	88	125	106	152	225	190	499	491	486	510	462	415	510	443	386
Jute	1043	1049	1186	1245	1833	2026	2329	2389	2396	2639	2549	2457	2639	2517	2467

*Include groundnut, rapeseed & mustard, sesamum, linseed, castorseed, nigerseed, safflower, sunflower and soyabean.

Source: Indian Economy, Mishra and Puri, 36th Edition, 2018 and Col 3-7 from Economic Survey, 2000-01 (Statistical Tables, Table 1.14), Col 7 to 16 from Economic Survey, 2019-20 (Volume II), Statistical Appendices, Table 1.17, and page number A 36.

Even if we focus on trends in agricultural growth in post-1991 economic reforms regime, it is observed that new economic policy seems to have bypassed the agriculture sector completely. The comparison in aggregate GDP growth vis-à-vis Agri-GDP growth is reflected in Table 13.3.

Table 13.3: Rates of Growth (in per cent) since 1991

Years	Overall GDP	Agri-GDP
1991-92 to 1995-96	5.2	2.4
1999-00 to 2003-04	6.0	2.9
2004-05 to 2013-14	8.0	3.7
2014-15 to 2018-19	7.2	1.9

Check Your Progress 1

- Trace out the trends in production and productivity of major crops in India after independence.

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- 2) Do you agree with the statement that ‘Agriculture has significantly contributed to development of India’?

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- 3) Do you agree with the view that agriculture growth continued to stagnate even after economic reforms were launched in 1991? Does it imply that economic reforms have by-passed agriculture?

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13.3 CAUSES FOR STAGNATION IN AGRICULTURE GROWTH IN INDIA

13.3.1 Deceleration in Agriculture Investment

Capital formation is necessary for improving long-term growth potential in agriculture. The share of agriculture and allied activities in gross capital formation in the economy increased in the mid-1960s and this trend continued till the late 1970s. Higher growth rates of agriculture witnessed in the 1980s were due to the lagged impact of increases in the share of agriculture and allied sector in gross capital formation during the late 1960s and 1970s (Figure 13.1)



Figure 13.1: Share of Agriculture and Allied Sectors (per cent) in Gross Capital Formation since 1966-67

Source: Planning commission of India and Agricultural Statistics at a Glance (2015)

However, since the 1980s, the share has shown a declining trend. There was a mild recovery during the late 1990s till 2001- 02, and then the share declined again. Total investment in agriculture as proportion of GDP declined from 9.9 per cent in 1990-91 to just 3.5 per cent in 1999-2000. The share of public

investment in gross capital formation in agriculture has also declined over the years, as shown in Table 13.4.

Table 13.4: Share of Public & Private Sector in Capital Formation in Agriculture and Allied Activities

Sector	1970s	1980s	1990s	2000s	2011-12	2015-16
Private	66.5	58.3	76.1	81.1	84.9	81.98
Public	33.5	41.7	23.9	18.9	15.1	18.02

Source: National Accounts Statistics and National Accounts Statistics, 2017.

There is a difference in the nature of public and private capital formation and contribution in the production processes. The former is mainly in the nature of public goods such as irrigation projects and road networks. These will not be provided by private capital. Decline in public capital formation is not adequately compensated by an increase in private investment in agriculture. The most important cause of decline in public investment in agriculture is the diversion of resources from investment to current expenditure in the form of subsidies.

13.3.2 Inadequate Irrigation Coverage and Falling Productivity of Irrigation in Foodgrains

Inadequate irrigation cover for most of the crops is an important constraining factor in speedy adoption of improved technology. Various Economic Surveys released by Ministry of Finance have highlighted the following issues: (a) only 40 per cent of Gross Cropped Area (GCA) was under irrigation in 2002-03, which increased only to around 49 per cent in 2016-17, and (b) share of public expenditure on irrigation and flood control to total public expenditure has declined over the years, (c) Irrigation coverage across the states is quite skewed (e.g. in Punjab 98.5 per cent of total area under all crops is irrigated, while in Maharashtra it was only 19.5 per cent in 2013-14) and (d) distribution of irrigation facilities across crops is equally skewed (e.g. 93.6 per cent of area under wheat, 60 per cent of area under rice has been under irrigation while for pulses and oilseeds it was only 20 per cent and 27 per cent, respectively).

Jha and Acharya (2011) examined trends in policies that govern public expenditure on agricultural and rural development, including expenditure on irrigation and flood control, in the post-Independence period. Their study classifies this period into three phases. The first phase, which lasted from the early years after independence to the late 1960s, saw the Central and State governments paying greater attention to the development of irrigation. This period was characterised by large outlays on medium and minor irrigation schemes. In the second phase, from the early 1970s to the late 1980s, substantial expenditure was incurred on new water technologies to promote the Green Revolution. The third phase, which begins with liberalisation of the economy in the 1990s, is characterised by a substantial reduction in the expenditure on agricultural and rural development. Table 13.5 shows the decline in the share of outlays on irrigation in GDP over time from the already low levels of the 1980s.